

Lecture Study Guide Addendum

This addendum provides a list of foundational study guide items, and it provides lists of correctly spelled terms for the spelling, formula, and equation items.

Foundational items

About 15% of the items on the study guide are designated as foundational items. These items address foundational topics. Compared to other items on the study guide, *foundational items have a higher probability of appearing on learning assessments*. These foundational items are listed below.

Definitions that can be given by writing lyrics to a song.

Question	term
15	mean
16	standard deviation
26	d -statistic
50	standard error of estimate
53	standard error of measure
64	standard error of a statistic
68	t -value
70	significance

Questions about types of distributions (The Googly Eyed Sheep Table)

Question	Type of distribution
45	Standard deviation
49	Standard error of estimate
52	Standard error of measure
56	Standard error of mean
58	standard error of a difference between means
60	standard error of a beta
62	standard error of a correlation

Equations where the average deviation of $|e|$ is a specific type of distribution:

Question	Type of distribution
44	Standard deviation
47, 48	Standard error of estimate
51	Standard error of measure
55	Standard error of mean
57	standard error of a difference between means
59	standard error of a beta
61	standard error of a correlation

Linear Equations for variance within a sample and variance with a population

Question	Statistic
72, 73	mean
94, 95, 96	difference between two means
119, 120, 121	Simple regression

Questions about key formulas

Question	formula
22	z-score
27	d-statistic
40	variance and covariance
43	correlation
117	slope
123	standard error estimate

Correct terms for spelling, formula, and equation items

The study guide includes some items that need to be typed and spelled correctly. This includes fill-in-the-blank spelling answers and items answered by providing formulas or equations. These questions are graded automatically by Canvas, and they will be marked as incorrect if they are not typed in a way that **exactly matches** an answer key. Although the Canvas answer keys will be set up to accept some common typos, the only way to guarantee an answer will be accepted by canvas is to type the answer precisely. You can recognize the fill-in-the-blank spelling questions because they will specifically ask you to “correctly spell” your answer. You can recognize the equation and formula questions because they will ask for an equation or formula.

Because these questions need to be answered precisely, you may want to check to make sure you have the correct spelling for each term, formula, or equation. The sections below provide a list of the correctly spelled terms, formulas, and equations that are needed for each unit. Thus, you can check the answers you derive for these items on the study guide with the lists provided below. If you derived an answer to a spelling, formula, or equation item that does NOT appear on the list, it means you do not have a correctly spelled answer.

Note that the items are sorted by unit, but within each unit they are NOT listed in a particular order.

Unit 1

Correct spelling terms

experimental, parameter, descriptive, z-score, inferential, reliability, validity, correlational, bias or biased, expected value, T-score

Formulas and equations

$\sum(x_i - \bar{x})^2$
 $ss/(n-1)$
 $\sqrt{ss/(n-1)}$
 $\hat{x}_i = \bar{x}$
 $x_i = \bar{x} + e_i$
 $(x_i - \bar{x})/sd$
 $10z + 50$
 $x_i - \bar{x}$

Unit 2

Correct spelling terms

small, true score, samples, medium, large, 5, 10, 15, 20, 25, 50, 55, 60, 65, 70, 75, cases, case, sample, sample mean, predicted value, assessments, assessment, population, population parameter, individual, normal.

Formulas and equations

prediction_s
 x_{is}
 $y_{is}=w+e_{is}$
 $(\bar{y}-\bar{y})_{sp}=(\mu-\mu)_p+e_{sp}$
 $r_{sp}=\rho_p+e_{sp}$
 $b_{sp}=\beta_p+e_{sp}$
 $x_{ai}=\text{true}_i+e_{ai}$
 $\bar{x}_{sp}=\mu_p+e_{sp}$
 $(\bar{x}_1-\bar{x}_2)/\text{sd}_{\text{pooled}}$
 $\sum(x-\bar{x})^2/(n-1)$
 $\sum((x-\bar{x})(y-\bar{y}))/((n-1))$
 $\text{cov}_{xy}/((\text{sd}_x)(\text{sd}_y))$
 $x_{is}=\bar{x}_s+e_{is}$
 $(\text{statistic}-\text{parameter})/\text{sd}_{\text{statistic}}$

Unit 3

Correct spelling terms

descriptive, inferential, variable, parameter estimate, probability

Formulas and equations

$8=3+(4)(1)+1$
 $4=0+4$
 $3=2+1$
 $3=0+3$
 $2=0+2$
 $x_{is}=\bar{x}_s+e_{is}$
 $y_{is}=\bar{y}_{0s}+(\bar{y}_{1s}-\bar{y}_{0s})(x_{is})+e_{is}$
or $y_{is}=\bar{y}_{0s}+(\bar{y}_{1s}-\bar{y}_{0s})x_{is}+e_{is}$
 $(\bar{y}_{1sp}-\bar{y}_{0sp})=(\mu_{1p}-\mu_{0p})+e_{1-0sp}$
or $\bar{y}_{1sp}-\bar{y}_{0sp}=\mu_{1p}-\mu_{0p}+e_{1-0sp}$
 $\bar{x}_{sp}=\mu_p+e_{sp}$
 $\bar{y}_{0sp}=\mu_{0p}+e_{0sp}$

Unit 4

Correct spelling terms

explained, error, total

Formulas and equations

$b_{1sp}=\beta_{1p}+e_{1sp}$
 $b_{0sp}=\beta_{0p}+e_{0sp}$
 $\sqrt{ss_{\text{est}}/(n-2)}$
 $\sum(y-\hat{y})^2$
 $\text{cov}_{xy}/\text{sd}_x^2$
or $\text{cov}_{xy}/((\text{sd}_x)(\text{sd}_x))$
 $y_{is}=b_{0s}+(b_{1s})(x_{is})+e_{is}$

Unit 5

Correct spelling terms

y, x_1 , x_2 , 0, 1, constrained

Formulas and equations

$$b_2$$

$$b/(a+b)$$

$$(b+c+d)/(a+b+c+d)$$

$$e_y.x2=b_0+(b_1)(e_{x1}.x2)+e$$

$$b/(a+b+c+d)$$

$$(b+c)/(a+b+c+d)$$

$$y=b_0+(b_2)(x_2)+e_y.x2$$

$$x_1=b_0+(b_2)(x_2)+e_{x1}.x2$$

Unit 6

Correct spelling terms

coding, between, within

Formulas and equations

$$(\bar{y}_{\text{group}}-\bar{y}_{\text{grand}})^2(n_{\text{group}})$$

$$(y-\bar{y}_{\text{group}})^2$$

$$(y-\bar{y}_{\text{grand}})^2$$

$$b_0+(b_1)(.33)+(b_2)(.5)$$